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FILE NAVIGATE TEXT CODE ANALYZE RUN

Code Control Task Refactor Code Issues Debugger Run Section Run and Advance Run to End

C:\Users\ADMIN\Documents\MATLAB

Files

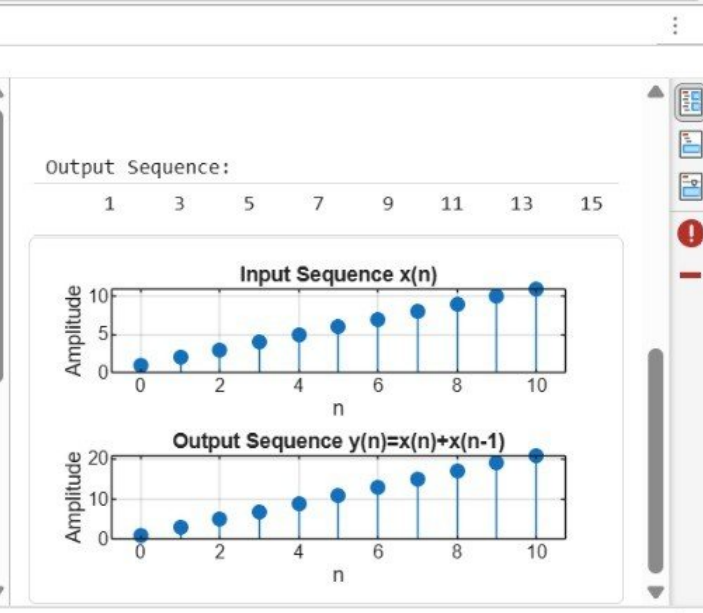
41.mlx
Addition and multiplication....
Discrete.mlx
Energy abd power 32.mlx
evenodd.mlx
Periodic.mlx
sinewave.mlx
triangleandsawtooth.mlx

Workspace

Name	Value	Size	Class
n	1×11 double	1×11	double
x	1×11 double	1×11	double
y	1×11 double	1×11	double

```

1 % Discrete-Time System
2 clc;
3 clear;
4 close all;
5 % Input sequence
6 n = 0:10;
7 x = [1 2 3 4 5 6 7 8 9 10 11];
8 % Discrete-Time System
9 y = filter([1 1],1,x);
10 % Display output
11 disp('Output Sequence:');
12 disp(y);
13 % Plotting
14 figure;
15 subplot(2,1,1);
16 stem(n,x,'filled');
    
```



Command Window

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MATLAB R2026a

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FILE NAVIGATE TEXT CODE ANALYZE SECTION RUN

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Files: 41.mlx, Addition and multiplication..., causal non causal 42.mlx, Discrete.mlx, Energy abd power 32.mlx, evenodd.mlx, Periodic.mlx, sinewave.mlx

Workspace:

Name	Value	Size	Class
n	1×11 double	1×11	double
x	1×11 double	1×11	double
y_causal	1×11 double	1×11	double
y_nonca...	1×11 double	1×11	double

Addition and multiplication.mlx x Energy abd power 32.mlx x 41.mlx x causal non causal 42.mlx x

C:\Users\ADMIN\Documents\MATLAB\causal non causal 42.mlx

```

1 % Causal and Non-Causal Systems
2 clc;
3 clear;
4 close all;
5 % Input sequence
6 n = 0:10;
7 x = [1 2 3 4 5 6 7 8 9 10 11];
8 % Causal System: y(n)=x(n)+x(n-1)
9 y_causal = filter([1 1],1,x);
10 % Non-Causal System: y(n)=x(n)+x(n+1)
11 y_noncausal = x + [x(2:end) 0];
12 % Plotting
13 figure;
14 subplot(3,1,1);
15 stem(n, x, 'filled');
  
```

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Input Sequence x(n)

Causal System Output y(n)=x(n)+x(n-1)

Non-Causal System Output y(n)=x(n)+x(n+1)

Amplitude

n

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